

MASTERS KERALA RE EXPO 2026

Comprehensive Features & Functionality Document

System Overview: ExpoKerala is a comprehensive event management platform built on Laravel 12 designed for Kerala's largest Renewable Energy, Battery Energy Storage Systems (BESS), and Electric Vehicle (EV) Showcase organized by Masters Association.

1. Technical Architecture & Stack

Component	Technology Stack
Backend Framework	Laravel 12.x (PHP 8.2+)
Frontend Build Engine	Vite, TailwindCSS, AlpineJS, FontAwesome
PDF Generation Engine	Barryvdh DomPDF (dompdf/dompdf v3.1)
Dynamic QR Code Engine	SimpleSoftwareIO QR Code Generator v4.2
Authentication & Profile	Laravel Breeze Middleware & Auth System
External Integration	Centralized Event API System (https://event.rowbest.in)

2. Key Features Breakdown

? A. Interactive Public Portal

- **Hero Banner & Event Countdown:** Event schedule for 9, 10, 11 January 2026 at Puthiyakavu Ground, Thripunithura, Ernakulam.
- **Exhibitor & Speaker Profiles:** Showcases 700+ MNRE-registered vendors, dignitaries, and event partners.
- **Floor Plan & Navigation:** Interactive map layout for visitor orientation.

?? B. Visitor & Sub-Event Registration

- **Online Onboarding:** Captures name, designation, company name, city, email, and mobile contact.
- **Dynamic Sub-Event Routing:** Supports custom registration pages via dynamic slug parameters (`/sub-event-register/{slug}`).
- **Remote API Synchronization:** Asynchronous HTTP POST validation with central backend databases.

? C. Real-Time PDF Badge & QR Code Generation

- **Instant Pass Generation:** Streams a personalized entry pass directly to the user browser upon registration.
- **256px Dynamic QR Code:** Encodes verification URL (`https://event.rowbest.in/register/{qr_id}`) rendered as inline Base64 data.
- **Unique Badge ID:** Padded 4-digit sequential identifier (e.g. `0042`).
- **Print-Ready PDF Output:** High-resolution formatting with company logos and venue details.

? D. Exhibitor Portal

- Custom registration workspace for clean energy suppliers, stall size requests, and category choices.

3. Application Route Summary

Route Path	HTTP Method	Controller / Purpose
/	GET	pageController@index (Main Landing Page)
/visitor-registration	GET	pageController@visitorsRegister
/visitors-register	POST	pageController@saveVisitorsRegister (API Sync)
/badge	GET	pageController@badgeSuccess (Generates PDF Badge)
/exhibitors-register	GET / POST	pageController@exhibitorsRegister
/sub-event-register/{slug}	GET / POST	pageController@subEventRegister
/dashboard & /profile	GET / PATCH	Laravel Breeze Auth & Profile Management